4weeks AI Course and Live Tutor Product Requirements

Founder build plan and commercial roadmap

Version 2 | 23 October 2026 | Arabic first | GCC launch

The launch decision

Build one useful course and one narrowly scoped live tutor before expanding the catalogue. The first product is AI for Everyone: a four week practical programme taught by your original AI human, with real browser demonstrations, assessed work and optional live help. The same lesson recipes and lab environment support recorded production and live instruction.

The first engineering proof is an eight minute recorded lesson and a ten minute live session in Arabic. The instructor must answer a course question, operate the Content Studio lab, verify its output, accept an interruption and hand control back. A polished face is only one part of that proof.

Decision	Specification
First market	Arabic speaking adults seeking practical AI skills for work or a small business. Start with one cohort and one country scope.
First course	20 lessons, four weeks, 40 learning hours, approximately five hours of recorded video and a portfolio project.
Original presenter	Founder creates the character, voice and video pipeline. No managed avatar subscription is required by this plan.
Demo scope	One learner at a time, one browser lab, push to talk, bounded tools, saved artifact, visible source references.
Initial technical allowance	Approximately SAR2,045 for the first course and incremental live demo, excluding unpaid founder labour, taxes and professional review.
Scale planning case	At 10,000 monthly active learners: about SAR19,823 infrastructure and SAR104,823 total monthly operating allocation before payment fees.

All budgets are editable assumptions, not supplier quotations or measured load results. The delivered UI prototype and backend starter are separate from the unconnected live speech and avatar pipeline. This PRD defines the acceptance gates required before claiming a working production tutor.

1 How to use this specification

Read first	Purpose
Sections 2 to 6	Product promise, learner journey, interface, architecture and the lab computer.
Sections 7 to 13	Create your human, record demonstrations, connect live conversation, memory and assessment.
Sections 14 to 19	The full first course and a production process you can repeat.
Sections 20 to 24	Build requirements, contracts, demo setup, tests and staged rollout.
Sections 25 to 28	Twenty further courses and reusable lab patterns.
Sections 29 to 34	Pricing, startup cash, monthly costs and runway.
Sections 35 to 41	Saudi requirements, audit evidence, operating controls, sale scenarios, next steps and implementation recipes.
References	Official technical, pricing and regulatory sources checked for this revision.

Companion deliverables

Deliverable	What is ready and what it proves
Interactive UI prototype	Arabic and English navigation, curriculum, local video preview, editable lab, scripted teaching sequence, rule based checks and local preferences. It proves the interaction design.
Backend starter	Local FastAPI test mode, scoped learner sessions and memory, typed tool actions, export and an optional model/browser integration path. The README is authoritative for implemented endpoints and setup.
Editable cost workbook	Usage, provider rates, labour, subscription conversion, taxes, fees and valuation sensitivities. Change assumptions before committing money.
This PDF and editable DOCX	The target product and phased build plan. Requirements labelled as future work are not claims about delivered code.

Status language is deliberate: “implemented” means present in a delivered artifact; “target” means an acceptance requirement; “allowance” means money reserved; “illustration” means a scenario. No actual learner revenue, retention, live avatar latency or acquisition offer is asserted.

2 Product promise and differentiation

4weeks.ai should sell a completed capability: a learner can produce, explain and improve a useful piece of work in four weeks. The AI human makes demonstrations approachable and support accessible. The defensible product is the combination of practical labs, a dependable tutor, credible assessment and evidence that learners improve.

Differentiator	How it appears in the product	Evidence to collect
Arabic teaching that works	Clear Arabic, common English interface terms explained, examples relevant to GCC work. Test accents and dialects instead of assuming parity.	Transcription error samples, comprehension ratings, successful tasks.
A tutor with hands and eyes	The learner sees real lab changes, can ask why, stop a step and take control.	Action success rate, verified states, time to first useful result.
Assessments with evidence	Artifacts, deterministic checks, visible rubrics, revision and appeal.	Before/after samples, grader calibration, appeal reversals.
Memory with learner control	Keep useful learning preferences and misconceptions; show and delete stored memory.	Resume success and deletion/isolation tests.
Maintained courses	Version lessons and dependencies; show last review date and update notes.	Broken lab rate, days to repair, affected lessons.
Original instructor identity	Consistent character and voice with a documented rights chain.	Asset register and reproducible production process.

What to promise at launch

“Build your first practical AI workflow in four weeks, with Arabic lessons, a practice lab and feedback on your work.” Offer the live tutor as a limited beta until measured reliability supports broader availability. Describe the presenter openly as an AI instructor; avoid implying that a human professional is attending every session.

Targets for a first paid cohort

- Recruit 20 to 30 adult learners with a specific work goal. Collect a baseline task before instruction.
- Target at least 70% completing the first useful artifact in week one and 60% completing the final portfolio. These are internal go or revise thresholds, not market benchmarks.
- Ask whether learners would pay the standard price after using the product. Track refunds and support time alongside satisfaction.
- Do not advertise “first in the Middle East” without a fresh, defensible market review. The product can stand out without an unprovable exclusivity claim.

3 Learner journey

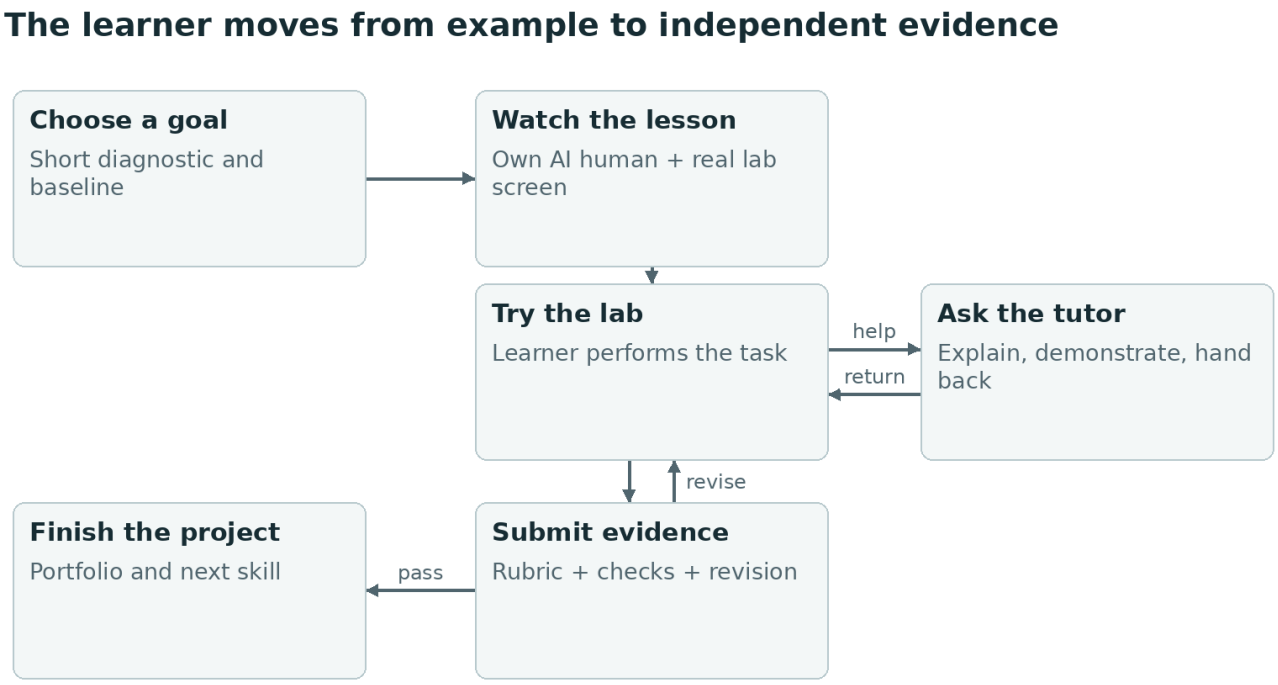


Figure 1 The same task moves through demonstration, practice, support and assessment.

Moment	Required behaviour
Onboarding	Choose a goal, Arabic preference and baseline task. Explain what the AI tutor can access and what it remembers.
Lesson	Show the learning outcome, video, transcript, source links, duration and last reviewed date.
Practice	Open a clean lab with a task fixture. Offer an example, hints and a reset without erasing submitted work.
Live support	Reserve minutes, warm the worker, show connection status, start a short session and keep Stop visible.
Assessment	Save the artifact, show criterion level feedback, request a revision and support an appeal.
Completion	Issue a completion record describing actual assessed skills and course version. Avoid unapproved accreditation wording.

4 Interface specification

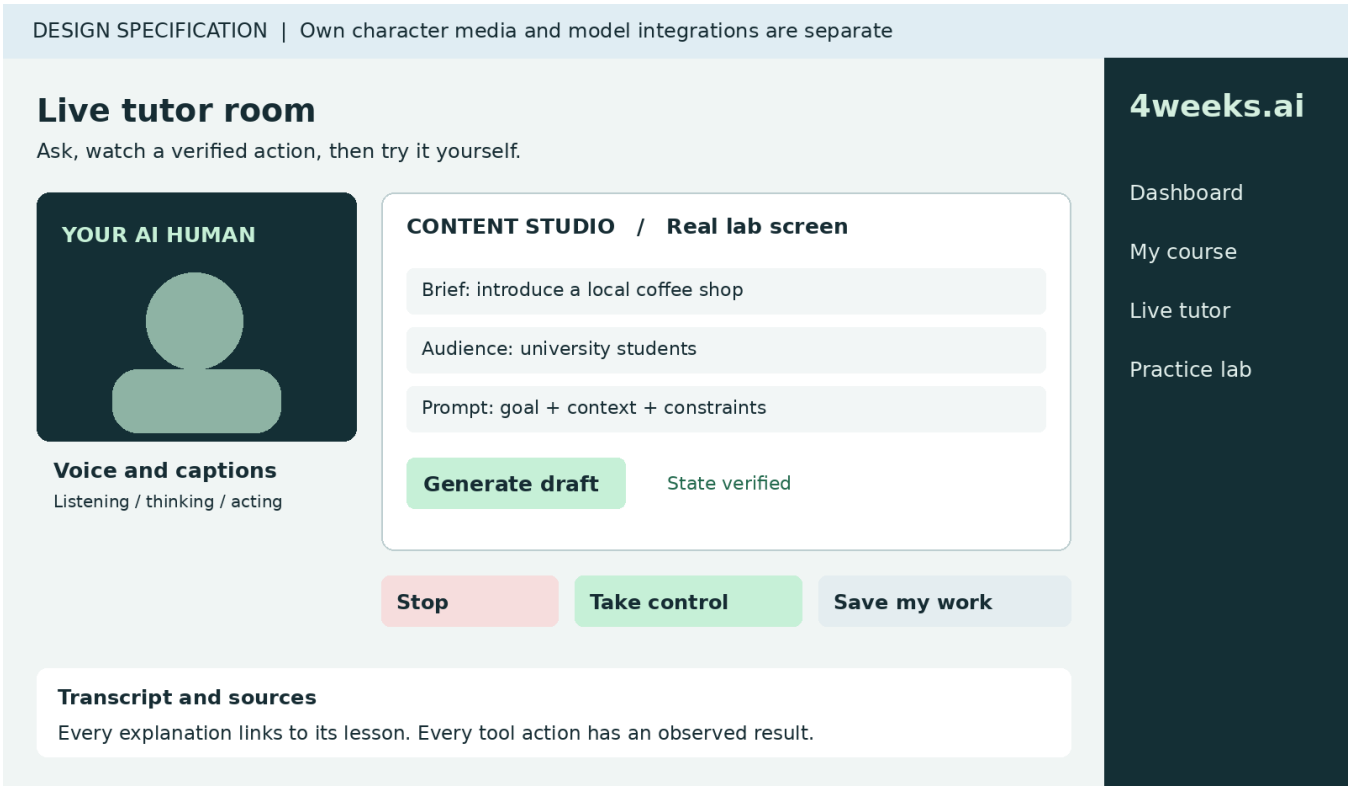


Figure 2 Target workspace layout. Character area is a design placeholder; actual presenter media is supplied by the founder.

Screen hierarchy

Use a calm workspace with a course navigation rail, a large lab screen, a smaller instructor area and a transcript below. Arabic is the default direction. Keep identifiers, URLs, code and timestamps readable in left to right islands. On mobile, use tabs for Instructor, Lab and Transcript; desktop is the preferred environment for complex labs.

Surface	Required controls and states
Dashboard and course	Continue lesson, weekly outcomes, progress, deadlines, saved artifacts and current support allowance.
Recorded lesson	Speed, captions, transcript search, large screen view, download permitted resources and “try this step”.
Live room	Join, warmup/queue message, microphone state, remaining minutes, Stop, Take control, Reset lab, reconnect and end.
Assessment and memory	Criterion evidence, revise, appeal, progress history, view memory and delete memory.
Admin	Course versions, failed labs, usage costs, grading review queue, incidents and publish rollback.

Use semantic buttons and visible focus, labelled form fields, captions, keyboard operation and contrast checks. A colour change alone must never communicate success or failure.

5 System architecture

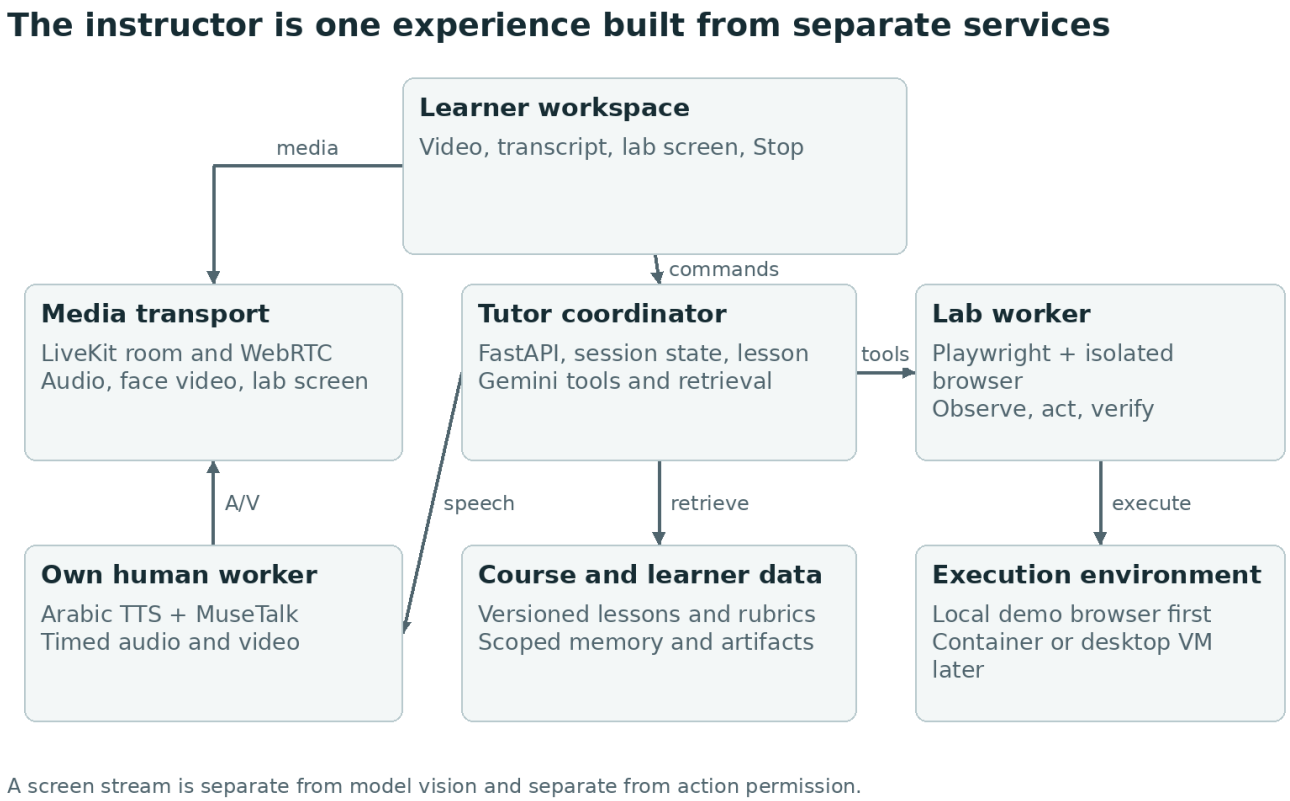


Figure 3 Separate media, reasoning, execution and learner data so each can be tested and replaced.

Layer	Selected starting stack
Web experience	React, TypeScript and Vite for the product. The delivered UI proof uses static HTML, CSS and JavaScript to remove build setup.
Coordinator	Python FastAPI; async jobs initially. PostgreSQL and a job queue when concurrency warrants them.
Reasoning	Gemini 3.5 Flash Lite via server API for questions, structured output and bounded tool decisions. It is a chosen model, not a claim to be the latest.
Speech and face	Whisper large v3 turbo through whisper.cpp; Chatterbox Multilingual V3; MuseTalk 1.5. Separate workers and compatible environments.
Realtime and lab	LiveKit WebRTC with a custom avatar worker; Playwright for controlled browser actions and screen capture.
Data and video	SQLite plus simple retrieval for the demo; PostgreSQL later. Object storage for artifacts and Cloudflare Stream for recorded lessons.

Gemini 3.5 Flash Lite supports the selected text, image and function calling role; its model page does not support Live API. The design therefore uses a separate speech recognition and synthesis pipeline. [S01]

6 The laptop and virtual lab

Your AI human does not need a physical laptop or a generated pair of hands that touches keys. The system operates a real software environment, captures its screen and presents it beside the human. A cinematic shot of your presenter at a laptop can frame a recorded lesson, but the instructional screen must come from the actual lab run.

Stage or subject	Environment	Reason and boundary
Founder demo	A dedicated local Chromium window controlled by Playwright. Founder shares that window.	Lowest cash cost. Use test data and a separate browser profile. The learner's personal computer is not controlled.
General browser courses	An ephemeral browser container or suitable managed sandbox per learner/session.	Reuse a template, scoped tool manifest and short lived credentials. Reset after the session.
Code labs	A sandbox with a tightly scoped filesystem and process policy.	Untrusted code needs a stronger isolation boundary than browser contexts alone.
Desktop software	A Linux or Windows desktop VM with screen transport and a separate action adapter.	Needed for real desktop applications. Account for OS/app licences, CPU/RAM, display streaming and image maintenance.
Image or video compute	A separate queued GPU job or metered external API.	Do not hold every learner's expensive video GPU idle during instruction.

Provision and dispose flow

- Authenticate learner and check entitlement. Create session and immutable lab template version.
- Provision the environment, attach only the lesson fixture, issue scoped credentials and perform a health check.
- Publish the screen. Read DOM or selected screenshots for model observation. Validate each proposed tool before executing it.
- Record action result, state version and artifact reference. Give the learner control when requested.
- Export permitted work, revoke tokens, stop the environment and expire temporary files. Test that a later learner cannot recover a previous learner's state.

Playwright browser contexts separate browser storage and cookies; this is useful session hygiene, not a complete security boundary for arbitrary native code. [S02]

7 Create the original AI human

Step	Founder action	Acceptance evidence
1 Character definition	Write a character sheet: face, age range, clothing, lighting, camera, posture and teaching manner. Generate an original identity or use explicitly licensed material.	Approved identity sheet and provenance register.
2 Base assets	Use an original portrait, for example FLUX.1 schnell. Create a clean short presenter base clip; optional Wan 2.2 TI2V 5B for gentle movement.	Consistent face, no unwanted identity resemblance, stable hands and background.
3 Voice	Create or license an Arabic voice. Test Chatterbox Multilingual V3 on Arabic and mixed Arabic/English tool names. Use owned or consented voice references.	Pronunciation list, reference consent and sample review.
4 Lip sync	Prepare face crop and precomputed assets for MuseTalk 1.5. Render approved speech against the base clip.	Readable mouth motion, no face drift and acceptable audio sync.
5 Production preset	Fix resolution, fps, loudness, camera crop, captions and instructor inset size. Keep templates versioned.	A repeatable five scene output with identical style.
6 Live adaptation	Warm a runtime worker and implement chunked audio, frame output, cancellation and room publishing.	Measured live demo; an offline render is insufficient evidence.

FLUX.1 schnell and Wan 2.2 TI2V 5B have Apache 2.0 model licences in their official cards; Wan’s published 5B path calls for at least 24 GB of NVIDIA VRAM with offload. Chatterbox and MuseTalk have their own code, model and dependency terms. Review the exact weights and every bundled component, not just the top level repository licence. [S03–S06]

Local hardware and cloud use

Use your existing Mac for the app, editing, local browser, speech experiments and orchestration. whisper.cpp provides Apple acceleration paths. Do not assume unified memory makes a CUDA project run unchanged. Rent NVIDIA compute for the components that require it, stop it after testing and persist only the needed models/assets. [S07]

MuseTalk changes the speaking face; it does not make a fully autonomous body or create arbitrary accurate hand actions. Full body movement is optional production work. Keep actual software operation independent of character animation.

8 Recorded course production

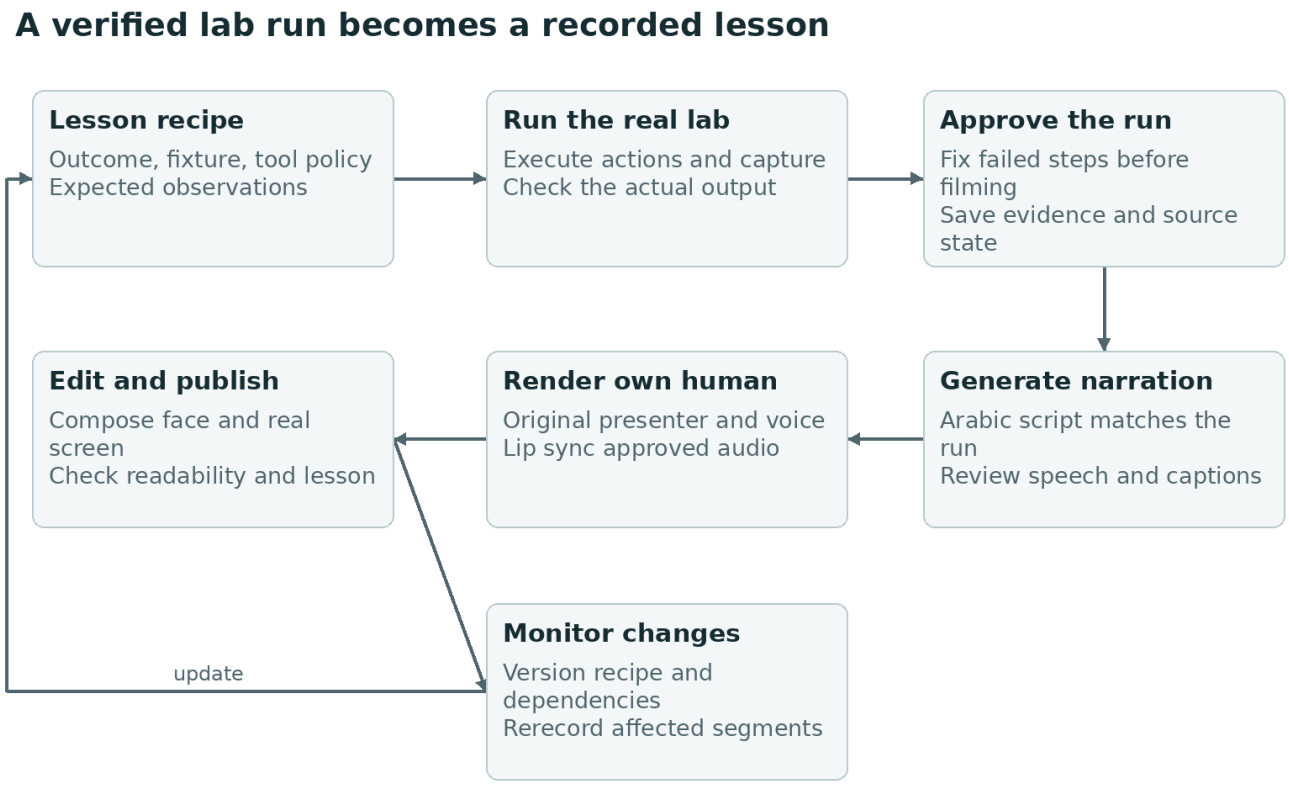


Figure 4 Narration is grounded in a verified screen run and the course is maintained by version.

Repeatable production steps

- Write the learner outcome, task fixture, solution rubric and expected state before scripting speech.
- Run the browser sequence with the actual model or app. Save inputs and outputs. Repair the demonstration until the steps are understandable and correct.
- Capture the real screen using a deterministic recording setup. Use readable text size and a steady cursor; hide unrelated tabs and credentials.
- Write or revise Arabic narration to match what happened. Generate voice, check names and numbers, then produce lip sync with your original presenter.
- Compose face and screen in an editor or FFmpeg. Use the face for explanations and make the screen dominant during actions. Add reviewed Arabic captions and chapter timestamps.
- Watch the complete export as a learner, perform the assignment from the published instructions, then publish the versioned lesson bundle.

OBS is an available open recording tool; production can use it or a scripted capture path. [S08] Keep separate source layers so a changed tool interface requires replacing a short screen segment instead of regenerating the entire human video.

9 The first recorded demo lesson

Lesson outcome: the learner writes a structured Arabic prompt for a small business, evaluates the generated result and improves it. The Content Studio lab contains business brief, audience, tone, constraints, prompt, generated draft and a save action. Use synthetic business details.

Time	On screen and instructor	Required evidence
0:00 to 0:45	State the outcome and show a weak prompt plus an unsatisfactory draft.	Visible baseline artifact.
0:45 to 2:00	Explain goal, context, audience, constraints and required output format.	Five elements highlighted in the actual prompt.
2:00 to 4:00	The instructor fills the lab and generates a draft through the real model adapter.	Action log and saved model output, not fabricated UI footage.
4:00 to 5:30	Check audience, factual claims, tone, length and call to action.	Rubric evidence, including one limitation or error.
5:30 to 6:45	Revise the prompt and generate again. Explain the change.	A second stored output and comparison.
6:45 to 7:30	Compare versions and export the better draft.	Downloadable artifact and clear reason for selection.
7:30 to 8:00	Give a different brief for the learner to solve independently.	Task fixture and completion criteria.

Example prompt structure

Act as an Arabic content editor. Write a short launch post for a fictional coffee shop serving university students. Use a friendly tone, state only the supplied facts, stay within 70 words and end with one clear call to action. Return the post and a list of claims that need verification.

Demo assets to prepare

- One original presenter reference and a short approved base clip; one licensed or owned voice reference.
- A clean lab template, baseline fixture, revised prompt, observed outputs and rubric.
- One eight minute narration script, voice track, screen capture, captions and final MP4.
- A manifest recording character, model, lesson, lab and rubric versions.

The delivered prototype accepts your local MP4 for review. It does not generate this final recorded lesson automatically, because your character and voice assets have not been supplied.

10 Live tutor runtime

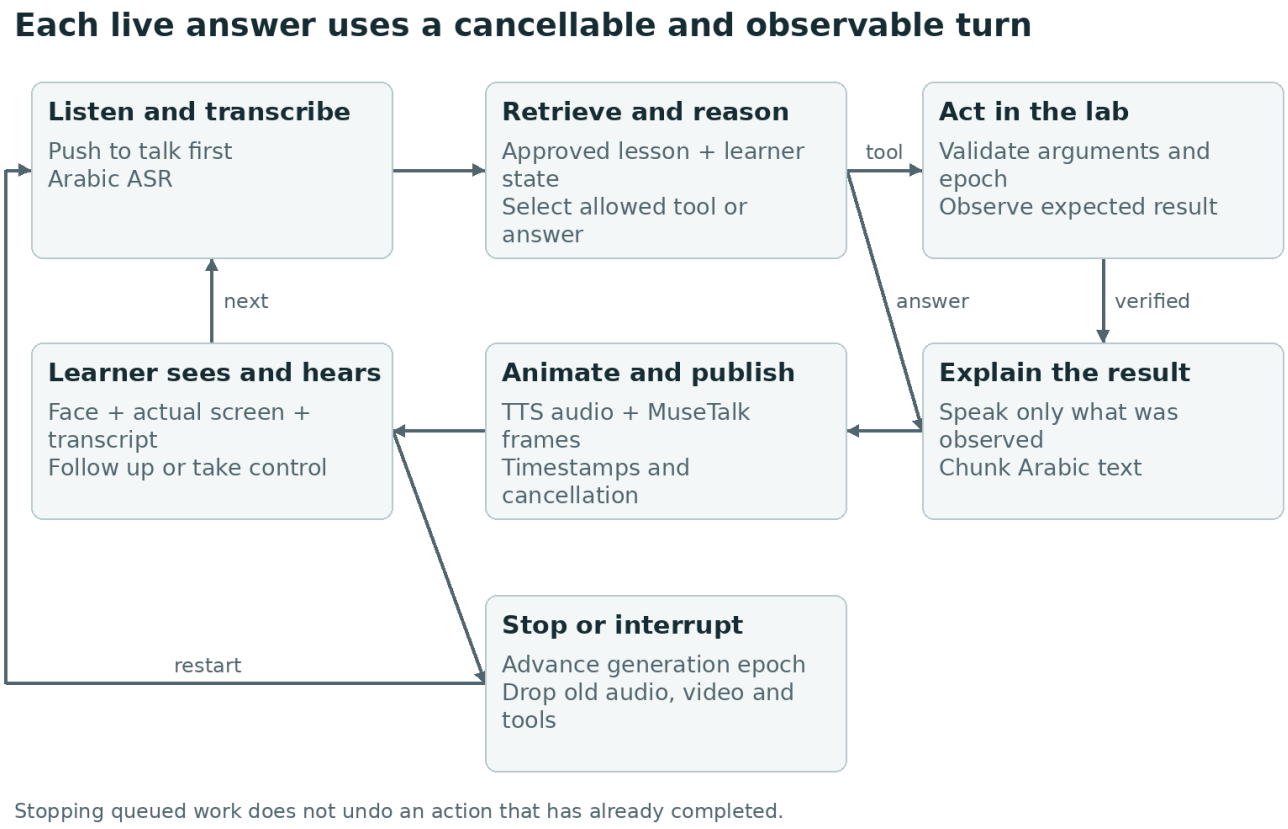


Figure 5 The live path joins conversation, grounded actions and timed audiovisual output.

The custom component you must implement

LiveKit supports custom avatar workers and an audio/video generator contract. Connect your worker to receive speech audio and publish synchronised audio/video frames. MuseTalk's published realtime inference example still reads an audio file; it is not a ready made conversational streaming server. [S09–S11]

Implement a warm worker that preloads character assets, accepts short speech chunks, preserves the model's necessary audio context, produces timestamped frames and can discard buffered work. A chunked sentence pipeline is a reasonable first experiment. Benchmark the complete path from learner turn end to audible response; model rendering speed alone is not conversational latency.

First live experience

Use push to talk and a ten minute session. Start with course questions and one lab action sequence. Add automatic turn detection and interruption after the baseline works. If the renderer can only produce short clips, label the experience as turn based while you improve streaming.

11 State and action control

State	Required user experience and system behaviour
Connecting and warming	Show progress, estimated queue position if available and Cancel. Do not start paid consumption before the session is usable.
Listening and thinking	Show microphone state and transcript. Record the current turn and retrieval sources.
Acting	Show the operation and target lab. Execute only validated tools; verify the new state.
Speaking	Publish audio and face frames with consistent timestamps. Show captions and source links.
Interrupted	Increase generation epoch, cancel pending speech/model work and discard old frames and queued actions.
Recovering	State the actual failure, preserve work, attempt a bounded retry and offer text or reset where appropriate.
Ending and ended	Save permitted artifacts and useful memory, close media, stop lab resources and settle measured usage.

Cancellation and correctness

Give each event a session_id, turn_id, generation_epoch and sequence number. Tool requests also carry an idempotency key. The Stop endpoint must work without waiting for the language model. Check the epoch immediately before any write and again before committing a delayed model result. Retrying the same save action must not duplicate the artifact.

Stopping speech cannot undo a click that already completed. Observe the current lab state and offer a defined undo or reset. Never say “I have saved it” until the save result and expected state have been observed.

Action return contract

Every action returns action_id, status, observed_state_version, structured result, screenshot reference when needed, duration_ms and an error category. An observation can be DOM state or a screenshot; the tutor should prefer structured state for known labs and use vision when the UI requires it.

Permission model

Start with open_lab, read_lab_state, set_brief, set_audience, set_tone, set_prompt, generate_draft, revise_draft, read_output, save_draft and reset_lab. Do not expose an unrestricted shell, arbitrary Python, free navigation or direct database access to the tutor. A new course adds a reviewed manifest of permitted actions.

12 Course knowledge and learner memory

Retrieval and memory solve different problems. Course retrieval supplies approved teaching content; learner memory supplies the learner's previous progress and preferences. Keep them separate so an incorrect conversation does not silently become an authoritative lesson.

Store	Minimum fields	Use
Approved content	course_id, lesson_id, section_id, version, language, text, source_url, reviewed_at	Retrieve small relevant passages and cite visible lesson sections.
Learner profile	server owned learner_id, language, explanation preference, consent/retention state	Tailor the explanation without unnecessary personal profiling.
Learning memory	misconception, last successful task, next practice, source_session_id, updated_at	Resume a useful teaching thread and allow correction/deletion.
Session history	turn id, transcript if retained, sources, tool results, usage, timestamps	Debug and provide continuity under an explicit retention policy.
Assessment record	artifact id, rubric version, criterion scores, evidence, reviewer changes	Explain why the learner passed or needs another attempt.

Demo retrieval

Start with 10 to 20 approved Arabic passages for the demo lesson. SQLite full text search or deterministic section lookup is sufficient to test grounding. Add multilingual embeddings and pgvector only when retrieval evaluation shows a benefit. Use a 10 question set containing answerable questions, ambiguous requests and questions outside the course.

Memory write policy

- Store a short structured learning summary after a session; do not append the entire conversation to every prompt.
- Let the learner view and reset memory. A removed memory must stop appearing in retrieval and future answers.
- Derive learner identity from authenticated server credentials. Never trust a learner id supplied by the model.
- Use separate test learners and intentionally attempt cross account retrieval before the pilot.
- Treat retrieved webpages and uploaded files as data. Embedded instructions must not override the tutor policy or tool permissions.

13 Assessment and feedback

Assessment should demonstrate a skill, explain the score and support improvement. A fluent model can produce convincing but inconsistent grades; combine objective checks with a rubric and review uncertain cases. Human quality review can remain part of operations even when every teaching video and tutor presentation uses your AI human.

Assessment type	Method	Escalation
Knowledge checks	Deterministic answers for objective questions; explanation after submission.	Flag ambiguous or outdated questions.
Tool execution	Verify saved artifacts, expected fields, formulas, output format or working code in the lab.	A broken lab cannot count as learner failure.
Creative or business work	Model assisted rubric with criterion level evidence and a bounded score.	Review low confidence, borderline passes and sampled submissions.
Capstone	Artifact, short rationale and a follow up task with changed inputs.	Appeal and founder or qualified reviewer check.

First course capstone rubric

Criterion	Weight	Evidence
Task and audience fit	20%	Clear goal, intended user and useful output.
Correctness and verification	25%	Claims checked, sources valid, calculations or logic inspected.
Prompt and workflow quality	20%	Reproducible steps, sensible constraints and revision rationale.
Artifact quality	20%	Readable, coherent, usable final work with suitable media.
Responsible handling and reflection	15%	Appropriate data, asset rights, limitations and explanation of choices.

Proposed pass rule: 70/100 with no unresolved critical factual or privacy failure. Calibrate against at least 30 founder reviewed samples before relying on automatic passing. Record disagreements and rubric changes; target agreement is an internal gate, not a claim that grading is unbiased.

Community and guidance

Use a small Discord community initially for weekly showcases, questions and peer feedback. Keep personal learner history and grades inside the platform. Hold one weekly group meeting during the first cohort to learn where the AI explanation fails. This can be founder facilitation and product research rather than a permanent tutor dependency.

14 First course structure

AI for Everyone is a practical adult beginner course. Learners need a laptop, basic browser skills and approximately ten hours a week. Define exactly which external accounts or paid tools are optional before purchase. The core lab should work without a separate paid subscription for every learner.

Week	Learning outcome	Portfolio result
1 Foundations and prompts	Use structured prompts, verify output and handle files responsibly.	Prompt pack and a verified short business draft.
2 Productivity and research	Produce a source checked brief and spreadsheet analysis.	Research brief, checked spreadsheet and one page memo.
3 Creative work and video	Plan, create and edit a short Arabic media piece with documented assets.	30 to 60 second video with captions and provenance.
4 Workflows and portfolio	Connect a small repeatable workflow, test it and defend the final project.	Working workflow and complete portfolio case study.

Activity	Per week	Four week total
Recorded lessons	75 minutes	5 hours
Guided and independent lab practice	6 hours	24 hours
Project and assessment work	2 hours	8 hours
Reflection and community	45 minutes	3 hours
Total	10 hours	40 hours

Twenty lessons average about 15 recorded minutes. Live help, when purchased, fits within practice time. The intensity comes from repeated practical work and feedback, not long passive videos. Use the same final project topic across weeks so tasks accumulate into a coherent result.

Final project choice

Choose one capstone track: work productivity, small business content, or learning and research. Each uses the same rubric and a repeatable workflow. Weekly practice still covers the full course; the final project develops one coherent use case rather than requiring three separate portfolios.

Weekly learning rhythm

Days 1 and 2: watch short demonstrations and complete guided exercises. Days 3 and 4: solve a new brief and ask for help only after attempting it. Day 5: submit the weekly artifact. Use the weekend for revision, the group review and recovery time. Keep a visible catch up path for working adults.

15 Week 1 Foundations and prompts

Weekly allocation: 75 minutes of recorded instruction, six hours of lab practice, two hours of assessment/project work and 45 minutes of reflection. Five sessions allow the learner to distribute work across the week.

Lesson 1 Getting started with AI

Outcome and lab: Select a small useful task, supply a clear request, and distinguish a generated draft from a checked final output. Open the course-owned Content Studio and inspect a fictional cafe brief. Deliverable: One saved request, one actual generated draft, and a three-item verification checklist.

Acceptance: The saved run contains a nonempty request, response, model identifier, timestamp, and fixture version. The final checklist contains at least three independently checkable requirements.

Lesson 2 How an AI assistant works

Outcome and lab: Use context deliberately and handle a question that the provided material cannot answer. Ask the same question with and without a supplied cafe information sheet. Deliverable: A comparison of two runs and one corrected uncertainty-handling instruction.

Acceptance: Both comparison runs use the same question and explicitly different context inputs. The learner explains, in one or two sentences, what information would resolve the uncertainty.

Lesson 3 Write a clear prompt

Outcome and lab: Construct an Arabic business prompt containing task, context, constraints, and output format. Fill the fictional business, audience, offer, and communication goal fields. Deliverable: A four-part Arabic prompt, the resulting post, and a checked rubric.

Acceptance: The prompt explicitly contains the task, audience/context, constraints, and desired output form. The learner verifies that the draft does not invent a new discount or business claim.

Lesson 4 Improve results through revision

Outcome and lab: Improve one output by changing a specific requirement while preserving verified facts. Load the L03 draft and ask for a version no longer than 35 words. Deliverable: Two versions, one revision instruction, and a brief change log.

Acceptance: The revised version contains no more than 35 whitespace-delimited words. The change log identifies at least two deliberate differences between versions.

Lesson 5 Week one challenge

Outcome and lab: Apply the prompting method independently to a different fictional business brief. Read a new fictional bookstore or training-event brief. Deliverable: A complete prompt-to-reviewed-output record for the new fixture.

Acceptance: The prompt references the new fixture rather than leaving cafe-specific fields behind. The submission includes the original output, revised output, and reasons for the changes.

Weekly submission: Prompt pack and verified business draft.

16 Week 2 Productivity and research

Weekly allocation: 75 minutes of recorded instruction, six hours of lab practice, two hours of assessment/project work and 45 minutes of reflection. Five sessions allow the learner to distribute work across the week.

Lesson 6 Writing and professional email

Outcome and lab: Draft a professional message appropriate to its audience and revise its tone without changing the facts. Generate a draft email requesting a meeting about a fictional project. Deliverable: A reviewed email draft and tone comparison; no email is sent.

Acceptance: Subject, greeting, purpose, proposed next step, and closing are present. No new promise, price, or commitment is added, and no outbound-send tool is invoked.

Lesson 7 Summaries and structured thinking

Outcome and lab: Convert a source document into an accurate summary and distinguish decisions from proposed actions. Load a supplied meeting transcript and request a short summary. Deliverable: A 150-word-or-shorter summary and an action table linked to source excerpts.

Acceptance: The summary is within 150 whitespace-delimited words and preserves all three seeded decisions. Unknown owners and deadlines remain marked unknown rather than being invented.

Lesson 8 Research and source checking

Outcome and lab: Answer a bounded question using traceable evidence and distinguish a source fact from an inference. Investigate a low-stakes product or learning-resource comparison using at least three relevant primary sources. Deliverable: A short research note and a claim-to-source table.

Acceptance: At least three non-duplicate source URLs or snapshot identifiers are present. The seeded misleading claim is corrected or explicitly marked unresolved.

Lesson 9 Spreadsheets and simple data

Outcome and lab: Clean a small dataset and verify a simple total or comparison independently of an AI explanation. Import a small fictional sales or study-time dataset. Deliverable: A cleaned CSV, one working formula or calculation, and one chart or summary table.

Acceptance: The cleaned file has the expected row count and numeric types. The written insight is consistent with the table and does not invent causation.

Lesson 10 Productivity challenge

Outcome and lab: Combine a document, a source check, and a small calculation into a useful reviewed deliverable. Read a new fictional project brief with an attached mini-dataset. Deliverable: A one-page memo with a verified calculation and evidence table.

Acceptance: The memo answers the assigned question and stays within the specified one-page structure. Every externally sourced factual statement has a valid evidence-table reference.

Weekly submission: Research brief, checked spreadsheet and one page memo.

17 Week 3 Creative work and video

Weekly allocation: 75 minutes of recorded instruction, six hours of lab practice, two hours of assessment/project work and 45 minutes of reflection. Five sessions allow the learner to distribute work across the week.

Lesson 11 Creating images with AI

Outcome and lab: Translate a visual goal into an image brief and review the generated image against explicit requirements. Write a brief covering subject, composition, palette, and intended use. Deliverable: One selected image, its generation brief, and a visual-review checklist.

Acceptance: The image file opens and matches the requested aspect ratio. The submission records the model, input assets, and intended usage; quality judgments receive rubric review rather than a fabricated automatic certainty score.

Lesson 12 Stories and scripts

Outcome and lab: Write a short script with a single outcome, a clear sequence, and visual instructions that can be produced. Choose a simple explanation or fictional campaign objective. Deliverable: A timed script and three-to-five-scene storyboard.

Acceptance: The storyboard contains three to five scenes with start/end timing and nonempty narration. The opening, explanation, and ending each support the same stated learning or communication goal.

Lesson 13 Arabic voice and narration

Outcome and lab: Generate or record clear Arabic narration and correct difficult names, numbers, and mixed-language terms. Split the approved script into short narration segments. Deliverable: An accepted narration file, transcript, and pronunciation notes.

Acceptance: The exported audio opens and its duration fits the storyboard after any documented timing revision. The transcript matches the intended spoken content and the voice-use basis is recorded.

Lesson 14 Creating and editing video

Outcome and lab: Assemble a short video with readable visuals, synchronized narration, and captions. Import the approved visual assets and narration into the real editor. Deliverable: One exported MP4 and its editable project or assembly manifest.

Acceptance: The MP4 plays for the planned 30–60 second duration with both video and audio tracks. No missing-asset placeholder or unintended black segment remains in the export.

Lesson 15 Short-video challenge

Outcome and lab: Complete and revise a short media piece from a new brief using the complete production checklist. Produce a short explanation for a new low-stakes topic or fictional business. Deliverable: A reviewed short video, source-asset list, and one documented revision.

Acceptance: The final video passes the playback, duration, caption, and missing-asset checks from L14. The submission includes the peer comment and a specific change made in response.

Weekly submission: Finished Arabic video and asset manifest.

18 Week 4 Workflows and portfolio

Weekly allocation: 75 minutes of recorded instruction, six hours of lab practice, two hours of assessment/project work and 45 minutes of reflection. Five sessions allow the learner to distribute work across the week.

Lesson 16 Building a practical workflow

Outcome and lab: Turn a repeated task into a defined sequence with inputs, checks, and outputs. Map one task into three to six steps. Deliverable: A workflow checklist or small recipe and its successful run record.

Acceptance: The workflow has three to six ordered steps with nonempty inputs and expected results. At least one failure condition has an explicit stop, retry, or human-review response.

Lesson 17 Evaluation and improvement

Outcome and lab: Use a rubric to identify a real weakness and verify that a revision improves it. Score two seeded outputs using the same four-criterion rubric. Deliverable: Two completed rubric forms and one evidence-backed revision.

Acceptance: Identify one specific error in the original output. Every criterion has a score and a specific supporting observation. The revised output resolves that error without introducing a new contradiction.

Lesson 18 Start your final project

Outcome and lab: Define a manageable project and produce a first version that can be evaluated. Choose one of the final-project tracks and define the target user and measurable output requirements. Deliverable: A project brief, artifact inventory, first version, and issue list.

Acceptance: The brief defines one target user and three to five observable success requirements. The issue list identifies remaining work rather than presenting an unchecked draft as final.

Lesson 19 Review your project

Outcome and lab: Review accuracy, usefulness, clarity, and reproducibility before final submission. Check claims, calculations, file playback, captions, and asset records relevant to the selected track. Deliverable: A reviewed project version, second-input run, and final checklist.

Acceptance: All mandatory checks for the chosen track are completed with evidence. No unresolved critical factual contradiction or broken required file remains.

Lesson 20 Present what you made

Outcome and lab: Explain the original problem, demonstrate the result, and describe the role and limitations of AI honestly. Prepare a three-to-five-minute walkthrough or equivalent short presentation. Deliverable: A final portfolio package, walkthrough, and personal next-step plan.

Acceptance: The walkthrough includes problem, process, result, evidence, and limitations. AI-generated components and human review are accurately described; the learner does not claim unsupported automation or independent verification.

Weekly submission: Portfolio, workflow and final explanation.

19 Course authoring and maintenance

Use a course factory with small versioned packages. The first course establishes templates; each new field adds specialist content, a lab template and an assessment contract. The runtime should not be rewritten for every instructor.

Package	Required contents
Course brief	Target learner, prerequisite, four week outcome, scope, risks, price hypothesis and reviewer.
Lesson recipe	Outcome, fixture, expected observations, allowed tools, demonstration steps, narration, sources and assignment.
Lab template	Image/browser/app version, resource limits, reset procedure, network policy, test data and credentials policy.
Media bundle	Original character asset version, voice rights, approved audio, screen capture, captions and final export.
Assessment bundle	Rubric version, deterministic checks, example passes/fails, grader prompt and appeal process.
Release manifest	Dependencies, last review date, known limitations, reviewer signoff and rollback version.

Create the first full course

- Draft all 20 outcomes and assignments before producing the full media set. Remove lessons that do not support the final portfolio.
- Complete one lesson end to end and use it as the production quality reference. Measure founder hours per finished minute.
- Produce in batches of five lessons. Validate each lab before recording its narration.
- Run an independent learner through each week; fix unclear instructions before expanding the next batch.
- Create completion records and publish only after assignments, captions, resource links and reset behaviour have been checked.

Keep content current

Review provider changes weekly and run lab smoke checks on a schedule appropriate to the dependency. A failed check opens a repair task with an affected lesson list. Label the lesson as under review if the core exercise is broken. Fix just the affected segment, update the release manifest and show a meaningful update note to learners. “Always current” must be an operating process with an owner and budget, not an absolute promise.

20 Functional requirements

ID and priority	Requirement	Acceptance
FR01 P0	Account and entitlement	Server checks course access and remaining allowance; private objects cannot be fetched across accounts.
FR02 P0	Recorded lesson	Video, captions, transcript, resources and progress work; lesson version is recorded.
FR03 P0	Lab launch and reset	Known fixture loads, health check passes, reset restores fixture and keeps submitted artifacts.
FR04 P0	Bounded tutor tools	Schema validation, allowed tool check and server scoped identity on every action.
FR05 P0	Live session lifecycle	Warm, join, push to talk, stop, recover and end have visible states; usage is measured.
FR06 P0	Observation and verification	Tutor cannot mark a tool step successful without an observed result.
FR07 P0	Assessment	Saved submission, rubric version, feedback, revise and review/appeal path.
FR08 P0	Memory	Learner can inspect and clear persistent memory; tenant isolation is tested.
FR09 P0	Usage and cleanup	Per session cost record, time cap, idle expiry, lab disposal and runaway budget stop.
FR10 P1	Course administration	Create version, run checks, approve release, publish and roll back.
FR11 P1	Payments and invoices	Webhook idempotency, refunds, course access changes and compliant invoice process.
FR12 P1	Community and cohort	Schedule, showcase links, report content and moderation ownership.
FR13 P2	Multiple subjects and presenters	Configurable knowledge, character and lab manifest reuse the same coordinator.
FR14 P2	Organisation accounts	Seat management, organisation permissions and aggregate learning reports.

P0 identifies a first usable product requirement, not a claim that the starter already implements it. Payments, production authentication, streaming speech/video and hosted sandbox provisioning remain build phases. The private UI prototype intentionally contains no real billing or credentials.

21 Data and service contracts

Entity	Required identifiers and fields
Course and lesson	course_id, version, lesson_id, language, outcome, media_ids, lab_template_id, rubric_id, reviewed_at.
Enrollment and entitlement	learner_id, course_id, payment/status, expiry, live allowance, organisation_id if applicable.
Tutor session	session_id, learner_id, course version, state, epoch, started_at, ended_at, worker/lab ids, usage totals.
Action event	action_id, session_id, turn_id, epoch, tool, validated args, idempotency key, before/after state, result.
Artifact and assessment	artifact_id, owner, storage key, content hash, submission_id, rubric version, evidence and review status.
Memory and knowledge	Learner scoped summaries; separate course scoped passages with source/version and deletion state.

Target API surface

Operation	Contract
Create session	POST /sessions: entitlement and template input; return session token, mode, lab status and expiry.
Ask or act	POST /sessions/{id}/messages or /actions: validated input plus current epoch; return turn/action id.
Observe	GET /sessions/{id}/state and event stream: state version, actions and permitted artifact references.
Stop and end	POST /sessions/{id}/stop and /end: advance epoch, close work, return final state and usage.
Memory and export	Read/reset memory; download only artifacts belonging to the authenticated learner.
Media token	Create short lived LiveKit tokens on the server after entitlement checks; never expose provider keys.

These are target contracts. The backend starter README lists the exact implemented route names. Keep identifiers opaque. Record UTC timestamps, currency and metering units. Store money in minor units and time in milliseconds or integer seconds; do not rely on floating point values for settlement.

22 Run the delivered demo

Try the interface now

- Open the private interactive prototype. Switch Arabic/English, enter the course and select a lesson.
- Load an MP4 you created into the recorded lesson preview. The file stays in your browser; it is not a published course upload.
- Open the live room and start the simulated session. Trigger the teaching sequence and observe the editable lab fields.
- Use Stop during the sequence, then Take control. Edit the prompt, generate the local template draft, inspect the rule based rubric and save a draft.
- Change the explanation preference, return to the dashboard and reset it. This demonstrates local preference UX, not RAG.

Run the backend on your computer

- Extract the backend starter ZIP into a new directory. Read README.md and .env.example before launch.
- Create a Python environment and install the declared dependencies and pin the resolved set after reproducing the tests. Start FastAPI using the README command.
- Keep deterministic mode for the first test. Create a session and run the included smoke/test commands. This exercises real server state without paid inference.
- Install the supported Playwright Chromium browser and enable its adapter when ready. Use the dedicated lab browser and synthetic data.
- Set your own Gemini key in the server environment to enable actual model generated drafts and responses. Never paste the key into the public frontend.
- Then add speech recognition, Arabic synthesis and LiveKit transport. Implement and benchmark the custom MuseTalk streaming adapter before claiming a live AI human.

What the starter does not supply

It is a local engineering starting point, not a production deployment. Your original character video/voice, cloud accounts, credentials and GPU are external inputs. Live avatar streaming, payments, production identity, full VM provisioning, real recorded course media and Saudi business approvals are separate work. The UI proof and backend starter are delivered separately; wiring the polished frontend to the API is an explicit integration task.

Keep an implementation status sheet in the repository. Every demo screen should distinguish real server behaviour, deterministic test behaviour and a simulated teaching sequence. Remove prototype labels only for capabilities that have passed their real integration tests.

23 Acceptance tests and measured gates

The following are proposed acceptance targets. Record hardware, model commit, configuration, sample count and p50/p95 for each run. A simulated UI test cannot satisfy an inference or streaming target.

Test	Target	Method
Arabic understanding	At least 18 of 20 representative utterances preserve the intended request.	Include Arabic/English tool names and selected GCC accents.
Grounded answer	At least 9 of 10 course questions correct with valid source references.	Include one unsupported question; accept an honest boundary.
Lab completion	At least 18 of 20 runs reach the expected saved state.	Changed inputs, slow network and one deliberately failing tool.
Warm response latency	Median at most 5s, p95 at most 10s from end of turn to first audio.	At least 30 real questions; report cold start separately.
Streaming	25fps sustained for 15min with no growing queue; p95 A/V skew at most 200ms.	Measure timestamps and buffer depth during speech.
Stop	Audio/video stop target at most 700ms; stale tools do not commit.	Interrupt a response and a pending model call; verify epoch.
Isolation	No learner A data returned to learner B.	Try memory, artifact, session and lab identifiers across accounts.
Recovery	Honest timeout/error state and preserved work.	Kill worker or disconnect network; retry only bounded actions.
Cost and cleanup	All resources expire and session costs reconcile.	Compare logs to provider usage; inspect orphaned workers.
Learning	A learner completes a changed task with less help.	Compare baseline to portfolio rubric and support use.

Release rule

Ship the recorded course when its lessons and assessments pass. Keep live support in a limited beta if the avatar path misses its latency or reliability targets. If the tutor works in voice and screen mode but the face lags, continue debugging the renderer without blocking recorded course sales or pretending the face is live.

24 Build phases and expansion gates

Phase and effort allowance	Build	Exit gate
A 2 to 4 focused days	Freeze the task, character design, lesson fixture, source passages, rubric and prototype flow.	A learner understands the task and completes the UI flow.
B 3 to 5 days	Run the real browser lab, capture the demonstration, generate approved narration and own human video.	One correct eight minute recorded lesson and assignment.
C 3 to 5 days	Connect backend, model, tools, scoped memory and server events to the frontend.	Real answers and saved lab artifacts; isolation and Stop pass.
D 5 to 15 days	Add ASR, Arabic TTS, LiveKit and custom streaming face worker.	Ten minute live session meets the measured acceptance tests.
E 3 to 5 weeks	Produce the full first course in batches and run a small cohort.	Completed portfolio evidence, price feedback and support cost data.
F Demand dependent	Add payments, production hosting, compliance, observability and a second course.	Paid launch gates and positive contribution economics.

These are focused founder work estimates, not a calendar promise. Custom avatar streaming is the largest uncertainty. Work can overlap, but course production and debugging still consume founder hours. Keep a short weekly review of finished outputs, blockers and measured costs.

Scale deliberately

Milestone	Required proof before increasing spend
20 to 30 pilot learners	The task is useful, the first course is finishable and failures can be recovered.
100 monthly active	One course operates reliably; know support minutes and cost per successful task.
1,000 monthly active	Several courses with repeatable QA, monitored concurrency and a paid acquisition experiment.
10,000 monthly active	Capacity tested against peak sessions, accountable support/content owners, reviewed financials and clean data/IP records.

25 Additional courses 1 to 5

These are 20 additional courses after AI for Everyone, for a 21 course catalogue. Launch order follows demand and shared lab cost. Every course still requires specialist review, a four week outcome and an assessed artifact.

1 AI for Arabic Content Marketing

Learner: Small business marketers. Capstone: Monthly content calendar with reviewed posts.

Lab: Browser. Action policy: Use approved brand facts; save drafts only. Assessment: Audience fit, evidence and useful variation.

Cost or review dependency: Marketing review and capped image credits. Suggested release: Wave 1.

2 AI for Video Production

Learner: Creators and marketing teams. Capstone: A finished short video with captions.

Lab: Browser plus queued GPU. Action policy: Generate short assets; approve publication separately. Assessment: Story, media quality, sync and rights.

Cost or review dependency: Generation credits and media review. Suggested release: Wave 2.

3 AI for Visual Design and Brand Assets

Learner: Design beginners. Capstone: A consistent campaign asset kit.

Lab: Browser plus image jobs. Action policy: Use versioned templates and bounded exports. Assessment: Layout, consistency and provenance.

Cost or review dependency: Image credits and design review. Suggested release: Wave 2.

4 AI for Arabic Writing and Translation Workflows

Learner: Writers and office workers. Capstone: Reviewed bilingual document and terminology guide.

Lab: Document browser. Action policy: Translate supplied text; preserve source meaning. Assessment: Accuracy, omissions and terminology.

Cost or review dependency: Language reviewer for specialist material. Suggested release: Wave 1.

5 AI for Sales Preparation

Learner: Sales teams and founders. Capstone: Research brief and outreach sequence.

Lab: Browser. Action policy: Synthetic prospects; no autonomous sending. Assessment: Relevance, truthful claims and privacy.

Cost or review dependency: CRM licences if selected. Suggested release: Wave 1.

26 Additional courses 6 to 10

These are 20 additional courses after AI for Everyone, for a 21 course catalogue. Launch order follows demand and shared lab cost. Every course still requires specialist review, a four week outcome and an assessed artifact.

6 AI for Customer Support

Learner: Support teams. Capstone: Knowledge assistant with escalation tests.

Lab: Browser. Action policy: Use synthetic tickets and scoped knowledge. Assessment: Resolution accuracy and escalation.

Cost or review dependency: Ticket licences and support reviewer. Suggested release: Wave 1.

7 AI for Ecommerce Operations

Learner: Store operators. Capstone: Product catalogue and store workflow.

Lab: Browser. Action policy: Operate a test store without real checkout. Assessment: Accurate listings and workable operations.

Cost or review dependency: Store licences and image credits. Suggested release: Wave 2.

8 AI for HR Administration

Learner: HR coordinators. Capstone: Policy assistant and onboarding workflow.

Lab: Browser. Action policy: Synthetic employees; no hiring decisions. Assessment: Privacy, policy accuracy and escalation.

Cost or review dependency: HR and legal review. Suggested release: Wave 2.

9 AI for Teachers and Course Designers

Learner: Educators and trainers. Capstone: Lesson and assessment pack.

Lab: Browser. Action policy: Use fictional learners; no student records. Assessment: Pedagogy, accessibility and factual accuracy.

Cost or review dependency: Qualified education review. Suggested release: Wave 2.

10 AI for Research and Evidence Synthesis

Learner: Researchers and students. Capstone: Literature brief and evidence table.

Lab: Browser plus notebook. Action policy: Inspect permitted sources and preserve citations. Assessment: Citation validity and methodological limits.

Cost or review dependency: Source access and research reviewer. Suggested release: Wave 3.

27 Additional courses 11 to 15

These are 20 additional courses after AI for Everyone, for a 21 course catalogue. Launch order follows demand and shared lab cost. Every course still requires specialist review, a four week outcome and an assessed artifact.

11 AI for Accounting Administration

Learner: Accounting teams. Capstone: Reconciliation and management report.

Lab: Browser or desktop VM. Action policy: Synthetic transactions; no financial execution. Assessment: Reconciliation and traceable assumptions.

Cost or review dependency: Qualified accountant and app licences. Suggested release: Wave 4.

12 AI for Legal Document Administration

Learner: Legal operations staff. Capstone: Document comparison and clause issue list.

Lab: Document lab. Action policy: Synthetic material; qualified review before reliance. Assessment: Traceability, confidentiality and uncertainty.

Cost or review dependency: Licensed legal reviewer. Suggested release: Wave 4.

13 AI for Healthcare Administration

Learner: Healthcare administrators. Capstone: Scheduling or documentation workflow.

Lab: Browser. Action policy: Synthetic records only; no clinical decisions. Assessment: Process accuracy, privacy and escalation.

Cost or review dependency: Qualified domain reviewer. Suggested release: Wave 4.

14 AI for Spreadsheets and Data Analysis

Learner: Office workers. Capstone: Clean workbook with verified formulas.

Lab: Browser or CPU desktop. Action policy: Use teaching workbooks and bounded file access. Assessment: Numerical correctness and reproducibility.

Cost or review dependency: Spreadsheet licences if selected. Suggested release: Wave 1.

15 AI for Business Intelligence and Data Storytelling

Learner: Analysts and operations staff. Capstone: Query backed dashboard and explanation.

Lab: Browser plus SQL sandbox. Action policy: Read only teaching database and bounded queries. Assessment: Query validity and honest interpretation.

Cost or review dependency: Database resources and analyst review. Suggested release: Wave 3.

28 Additional courses 16 to 20

These are 20 additional courses after AI for Everyone, for a 21 course catalogue. Launch order follows demand and shared lab cost. Every course still requires specialist review, a four week outcome and an assessed artifact.

16 AI for Presentations and Business Communication

Learner: Business professionals. Capstone: Evidence backed presentation and speaker notes.

Lab: Browser. Action policy: Use supplied facts and controlled exports. Assessment: Argument, readability and factual support.

Cost or review dependency: Presentation tool licences. Suggested release: Wave 2.

17 AI for Product and UX Design

Learner: Product builders. Capstone: Research synthesis and tested prototype.

Lab: Browser. Action policy: Synthetic research and scoped design tools. Assessment: Evidence, usability and design rationale.

Cost or review dependency: Design tools and UX reviewer. Suggested release: Wave 2.

18 AI for Software Development

Learner: Technical beginners. Capstone: Working small application with tests.

Lab: Code sandbox. Action policy: Ephemeral files and scoped preview deployment. Assessment: Functionality, robustness and security basics.

Cost or review dependency: Sandbox isolation and code reviewer. Suggested release: Wave 3.

19 AI for No Code Automation

Learner: Operators and founders. Capstone: Workflow with retries and execution log.

Lab: Browser plus worker. Action policy: Allowlisted integrations and fake credentials. Assessment: Correctness, recovery and cost.

Cost or review dependency: Execution quotas and connector limits. Suggested release: Wave 3.

20 AI Agents and Retrieval Systems

Learner: Technical builders. Capstone: Grounded assistant with bounded tools.

Lab: Code and retrieval sandbox. Action policy: Scoped knowledge, tool schemas and test accounts. Assessment: Retrieval quality, tool safety and evaluation.

Cost or review dependency: Model usage and engineering review. Suggested release: Wave 3.

Reuse the same session, media, memory, grading and billing services. Add a course manifest, role specific knowledge, lab image, fixtures, tool adapter and calibrated rubric. A different presenter identity is optional and should not require a different platform.

29 Prices and commercial packaging

Start with a one time course purchase. Introduce a membership after three to five useful courses exist and learners have a reason to return. The proposed prices below are experiments to test willingness to pay, not a claim about a market clearing price.

Offer	Proposed consumer price	What is included
Founding first cohort	SAR299	First course, labs and assessments; explicit beta terms and limited seats.
Standard four week course	SAR499	Structured course, portfolio assessment, completion record and defined access period.
Guided course	SAR799	Standard package plus 120 live tutor minutes over four weeks and weekly group guidance during the pilot.
Later membership	SAR149 per month	Course library, self practice and 30 live tutor minutes per month. Assessed cohorts or specialist reviews can remain separately priced.

For modelling, consumer prices include 15% VAT where applicable. Define access duration, renewal, included minutes, refunds and external tool requirements before checkout. Do not offer unlimited live use until the distribution of actual usage supports it.

Test pricing with real buyers

- Show one complete lesson, a real learner artifact and a brief live demonstration. State the time commitment and exact deliverables.
- Recruit the initial cohort through your own audience and relevant professional communities. Do not imply partnerships or credentials that do not exist.
- Track visit to trial, trial to paid, activation, completion, refund, repeat purchase and support burden by cohort.
- Interview non buyers and refunded buyers. Change the offer or outcome before simply lowering price.
- Add a team package only after multiple learners from the same organisation ask for reporting, billing or controlled access.

Unit economics rule

For each offer, subtract taxes, refunds, payment fees, delivered live usage, lab usage, grading and variable support from collected cash. Compare the remaining contribution with acquisition cost. One time course sales are revenue, but they are not annual recurring revenue. Distinguish recurring subscribers from purchasers with continuing access.

30 Cheapest technical kickoff

Item	Central allowance in SAR	Interpretation
Domain allowance	600.00	Budget for a new .ai purchase; ownership, availability and checkout price unconfirmed. Set to zero if already purchased.
Recorded production GPU	442.50	200 powered on hours at USD0.59 per hour. A spending allowance, not measured rendering throughput.
Recorded storage and APIs	240.00	100 GB GPU storage for two months: SAR52.50. Model API allowance: SAR187.50.
Prelaunch video and assets	112.50	Two months of Stream storage: SAR37.50. Optional licensed assets: SAR75.
Recorded contingency and total	279.00 / 1,674.00	20% of SAR1,395 subtotal. Local app/editing software and unpaid founder time add no cash here.
Incremental live demo GPU	132.75	60 powered on hours at USD0.59 per hour; includes setup, testing and idle time.
Incremental GPU storage	13.13	50 GB for one month at USD0.07 per GB.
Model API allowance	75.00	Budget cap for development and demonstration calls.
Live demo contingency	150.00	Allowance for reruns and minor technical costs.
Incremental live demo total	370.88	Local app/browser, one concurrent learner, transport within free allowances.
Combined technical allowance	2,044.88	Before tax, founder labour and independent review.

The recorded allowance replaces the earlier unitemised figure. Its two months of storage cover production before the operating schedule. The incremental demo is roughly SAR305 at 30 GPU hours or SAR459 at 100 GPU hours. These are spending allowances, not proof that the required visual quality and latency can be achieved at those hours. Do not purchase a permanent GPU or duplicate a domain already owned. USD1 = SAR3.75 is the budget conversion assumption. [S12–S13]

Before a paid public launch

Reserve approximately SAR2,000 to 5,000 for a bounded Saudi legal/privacy/accounting review if required for your entity and offer. A focused security review and retest may need a further SAR3,000 to 8,000 at broader release. These are founder planning reserves, not quotes; government fees, entity formation, remediation and complex reviews are additional. Deduct any work already covered by the monthly professional reserve to avoid double counting.

Why the earlier budget was much higher

The previous one course pilot envelope of SAR100,000 to 200,000 was not a bottom up minimum for your DIY demo. The broader 20 course central model included SAR270,000 of paid labour, SAR150,000 of other cash and SAR84,000 contingency, plus a separate operating buffer. This plan starts with your unpaid work, one lab, one learner and scheduled compute. It still recognises later content, support, security and acquisition costs.

31 Runtime assumptions through 10000 active learners

Monthly active learners are distinct people who perform a defined learning activity in a month. They are not the same as registered accounts, paying subscribers or simultaneous tutor sessions. Capacity and valuation must keep those counts separate.

Driver	100 active	1000 active	10000 active
Courses available	1	5	21
Recorded minutes per course	300	300	300
Viewing minutes per active learner	120	120	120
Live adoption	25%	25%	25%
Live minutes per adopter	30	30	30
Independent lab minutes per active learner	30	30	30
Concurrent live sessions per GPU	1	1	1
Effective GPU occupancy	60%	60%	60%
Warm GPU floor per month	0 hours	730 hours	730 hours
GPU storage	100 GB	100 GB	500 GB
Browser startup and idle multiplier	1.2	1.2	1.2
Peak to average concurrency	4 times	4 times	4 times

Provider basis

The model selects A6000 Secure Cloud compute at USD0.59/hour, E2B 2 vCPU plus 4 GiB browser compute at USD0.1656/hour, Cloudflare Stream at USD5 per 1,000 stored minutes and USD1 per 1,000 delivered minutes, and Gemini 3.5 Flash Lite at USD0.30 per million input tokens and USD2.50 per million output tokens. Rates were checked on 23 October 2026 and can change. [S12–S16]

LiveKit transport uses Build for the small pilot and Ship pricing in larger scenarios. The own avatar worker runs on the rented GPU, so the model does not add a LiveKit cloud agent deployment fee. Metering assumptions, bandwidth, regions and actual invoices must be verified. [S17]

Desktop VM licences, premium third party course apps, expensive image/video generation, full session recording and enterprise regional commitments are excluded from this browser based baseline. Add them by course before selling an all inclusive package.

32 Cost formulas and capacity

Driver	Formula or allowance
Live session hours	MAU × 25% live adoption × 30 minutes ÷ 60.
Billed GPU hours	MAX(live session hours ÷ sessions per GPU ÷ 60% occupancy, warm GPU floor).
Browser billed hours	(Live session hours + MAU × 30 independent practice minutes ÷ 60) × 1.2.
Recorded delivery	MAU × 120 minutes ÷ 1,000 × USD1.
Recorded storage	CEILING(courses × 300 minutes ÷ 1,000) × USD5 per month.
Live transport bandwidth	2 Mbps for live sessions; 1 Mbps for independent lab screen. Meter the actual bitrate.
Model usage	6,000 input and 600 output tokens per live minute, plus 20,000 input and 2,000 output tokens per MAU for other activity.
App services	USD50 / 150 / 500 monthly allowances for app, database, email and monitoring.

The 10000 active learner scenario

The selected workload produces 75,000 live minutes, 7,500 billed browser hours, 375,000 billable end user transport minutes and 3,375 GB of downstream realtime media per month. A four times average peak assumption implies about seven simultaneous GPU sessions and 42 browser labs. E2B Pro is included for that browser concurrency. [S14, S17]

The GPU formula assumes workers can scale with usable occupancy and enough available supply. One warm worker does not guarantee all seven peak sessions start instantly. Measure startup time, reserve capacity if necessary and expose a queue. A fully reserved seven GPU pool costs more than the demand based calculation.

Sensitivity and budgets

More live usage increases GPU, browser, transport and model costs together. Higher resolution raises bandwidth. Long prompts and chat history increase input tokens. First optimise failed attempts, idle resources and long answers, then test GPU concurrency. Never assume two sessions per GPU merely because the model fits in memory.

Log cost per delivered live minute, per successful lab task and per finished recorded minute. Add per session time caps and alerts at 50%, 80% and 100% of a spending allowance. A monthly user count alone is not enough to estimate operating cost.

33 Monthly operating budget

Monthly SAR	100 active	1000 active	10000 active
Avatar GPU compute	46.09	1,615.13	4,609.38
GPU storage	26.25	26.25	131.25
Recorded video	63.75	487.50	4,631.25
Browser labs and plan fees	46.58	465.75	5,220.00
Realtime transport	0.00	226.88	2,015.63
Language model usage	13.41	134.06	1,340.63
App and database allowance	187.50	562.50	1,875.00
Infrastructure total	383.58	3,518.06	19,823.13
People content and support	0	12,000	55,000
Marketing	0	5,000	25,000
Professional accounting audit reserve	500	2,000	5,000
Operating allocation before payment fees	883.58	22,518.06	104,823.13
Founder replacement for earnings analysis	10,000	15,000	20,000

Founder replacement is a separate economic normalisation and is not added to the cash operating row when the founder remains unpaid. Cash people/content/support allowances are planning allocations, not salary quotes. Assign actual owners and reconcile both founder and paid contributions before using profit in a sale discussion.

What the 10000 stage pays for

The SAR55,000 people allocation must cover some mix of engineering/operations, content maintenance, learner support, grading review and specialist input. If course complexity or service expectations require more staffing, increase it. The SAR25,000 marketing allocation is not evidence that 10,000 active learners can be acquired for that amount.

Infrastructure tax treatment, payment processing, refunds, corporate tax/zakat, financing, bespoke enterprise work and acquisition transaction fees are not included in this operating total. They appear in separate model inputs where applicable or need entity specific treatment.

34 Runway and growth cash

There is no defensible single cost “until 10,000 active students” without a timeline, acquisition funnel, paid conversion and retention data. The workbook therefore separates a monthly workload scenario from a cumulative funding schedule. Change the time spent at each stage and avoid presenting that schedule as a forecast.

Illustrative schedule only	Months	Monthly operating allocation	Stage spending
100 active workload	2	SAR883.58	SAR1,767.15
1000 active workload	4	SAR22,518.06	SAR90,072.25
10000 active workload	6	SAR104,823.13	SAR628,938.75
Technical kickoff	One time	SAR2,044.88	SAR2,044.88
Gross scheduled spending	12 months	Before receipts and extra one offs	SAR722,823.03

This example assumes those workloads already exist at each stage. It does not show how they are acquired. It includes the monthly marketing and professional allocations, so do not add the same expenditure again as separate CAC or audit costs. Additional course production, licensing, legal setup or security projects should be added only when they are outside those allocations.

Funding formula

Required external cash is the peak cumulative deficit after actual cash receipts, refunds, taxes, processing costs and operating payments, plus a chosen reserve. It is not automatically equal to total spending. A profitable later stage does not fund an earlier deficit unless the business survives long enough to receive that cash.

Retention and acquisition example

With 3,000 subscribers and 5% monthly subscriber churn, approximately 150 replacement subscribers are needed every month to stay level. A SAR25,000 marketing budget supports at most SAR166.67 acquisition spend per replacement subscriber. Net growth requires additional acquisition or better retention. These are arithmetic illustrations, not observed CAC or churn.

Track new paying users, cohorts, cancellation reasons, contribution margin and payback. Use low cost direct founder sales and a first cohort to learn before committing a large growth budget. Do not use free registrations as proof of future subscription revenue.

35 Saudi launch and audit preparation

Confirm the legal route for your actual entity, training offer and certificates before paid public launch. NeLC announced a unified private sector e learning licensing route in March 2026, effective May 2026, with private training programme licences issued through TVTC after FutureX technical integration. Avoid relying on an old blanket assumption that every offer needs the same two separate licences. [S18–S19]

Workstream	Concrete output	Timing
Training and certificates	Short applicability memo, responsible authority, required licences/integration, allowed certificate wording and renewal owner.	Before paid launch.
Privacy and transfers	Data inventory, legal basis/notice, retention and deletion, processor agreements and transfer assessment where applicable.	Before real learner data flows to providers.
Tax and invoicing	VAT applicability, compliant invoice process, payment reconciliation, refund/credit note treatment and accounting owner.	Before taxable sales at the relevant obligations.
Original assets and code	Company ownership/assignment, voice consent, model and dependency licences, source register and contractor terms.	Before production and whenever assets change.
Independent technical review	Defined security scope, report, fixes and retest evidence.	Before wider paid use and after material changes.

Saudi data rules require attention to actual processing and international transfers. A Saudi database alone does not resolve overseas GPU, model, analytics or logging processing. Map accounts, voice, transcripts, screens, assignments and embeddings to each provider and location; confirm the applicable safeguards. [S20]

ZATCA describes mandatory VAT registration at SAR375,000 annual revenue and voluntary registration above SAR187,500 for the individuals covered by that service; entity, residency and taxable supply details still matter. E invoicing has generation and integration phases with specific applicability. A payment gateway receipt is not automatically a compliant tax invoice. [S21–S22]

The professional reserves in this plan are planning amounts. They do not make the project “audited”. Record who reviewed what version, the date, scope, exceptions and remaining remediation.

36 Buyer ready audit evidence

Area	Maintain from the start	What a buyer can verify
Ownership	Company records, cap table, founder/contractor IP assignments, domain and vendor ownership.	The seller can legally transfer the product and its assets.
Financial	Monthly P&L, bank and gateway reconciliation, invoices, refunds, taxes and normalised earnings bridge.	Revenue is real, costs are complete and profit survives founder replacement.
Customer metrics	Registered, active, paying and recurring counts; cohorts, churn, retention and acquisition costs.	Demand quality without double counting free users or course purchasers.
Learning quality	Rubrics, calibration samples, completion and portfolio evidence, appeals and version history.	The product teaches measurable skills and handles errors.
Technical	Architecture, deployment instructions, tests, backups, restore drill, monitoring and incident records.	The platform can be operated and transferred without one person’s memory.
AI operations	Model/weight versions, prompts, evaluation sets, per task costs, failure rates and rollback.	Reliability and margins persist after model or provider changes.
Security and privacy	Access reviews, isolation tests, external findings/retest, data map, retention and deletion evidence.	Known risks are documented and materially remediated.
Compliance and licences	Training memo/approvals, FutureX applicability, tax process, content and software licences.	The business is operated within its actual permissions and obligations.

Data room structure

Use folders for corporate, financial, customer metrics, product, technology, intellectual property, security/privacy and legal/regulatory. Share aggregated or redacted learner evidence first. Do not export students’ personal tutor memories into a buyer data room merely to demonstrate engagement.

Audit scope and costs

Use a qualified accountant for financial work and an appropriately scoped security reviewer for technical controls. SOC 2 or ISO certification is not a universal prerequisite for a course demo; commission it when buyer or enterprise requirements justify it. Reserve roughly SAR15,000 to 40,000 for bounded sale readiness and diligence preparation, excluding success fees, taxes, major remediation and unusually complex reviews. This is a planning reserve, not a quote.

37 Operating quality and resilience

Control	Minimum operating practice
Content changes	Dependency watch, failed lab queue, named owner, lesson version and rollback. Review regulated field content with a qualified specialist.
Model changes	Pin tested versions when possible; rerun language, tool and grading evaluations before promotion.
Availability	Measure worker warmup, GPU capacity, lab startup and disconnections. Offer queue, retry or recorded learning fallback.
Access and secrets	Server side keys, scoped tokens, least privilege, rotation and no credentials in screenshots or client bundles.
Isolation and uploads	Per learner storage, file validation, size limits, malware handling where relevant and sandboxed execution.
Abuse and spending	Time quotas, rate limits, bounded prompts/tools, per account budget and administrative kill switch.
Backups	Back up durable course/customer state; test restoration. Ephemeral labs can be disposable, submitted artifacts cannot.
Support and incidents	Clear contact path, escalation owner, incident timeline and learner notification process when needed.

Metrics dashboard

Show activation, completion, assessment improvement and repeat purchase alongside live answer latency, tool success, interruptions, crashes, lab cleanup and cost. Segment Arabic recognition and feedback quality by tested language patterns. Use incident counts and failed tasks to prioritise work instead of only session duration.

Limits on 24 hour availability

“Available 24/7” needs a warm capacity policy, monitoring and an operational response. Scheduled beta windows are cheaper and more honest while you are the only operator. Upgrade that promise only after you have measured queue performance and arranged support for outages. The 730 hour warm floor in the larger model is a budgeting assumption, not an uptime guarantee.

38 Valuation and selling the business

Ten thousand active learners do not establish a sale price. A buyer will examine durable earnings or recurring revenue, retention, customer concentration, growth quality, technology ownership, transferability and risk. A platform with free users and no repeatable profit can be valued very differently from a profitable subscription business.

Illustrative 10000 active scenario	Monthly SAR
3000 subscribers at SAR149	447,000.00
After assumed 3% refunds and 15% VAT extraction	377,034.78
Payment fees at 2.5% of original billings	11,175.00
Net after those items	365,859.78
Operating allocation	104,823.13
Profit before founder replacement	261,036.66
Founder replacement normalisation	20,000.00
Normalised operating profit	241,036.66

Annualised normalised operating profit in this illustration is SAR2,892,439.89. This is not observed performance. One time course sales default to zero in the scenario to avoid counting the same access twice. Corporate tax/zakat, financing and buyer specific adjustments require separate treatment.

Annual sustainable profit	3 times	3.9 times	5 times
SAR300,000	SAR0.90m	SAR1.17m	SAR1.50m
SAR1,200,000	SAR3.60m	SAR4.68m	SAR6.00m
SAR2,892,440 illustration	SAR8.68m	SAR11.28m	SAR14.46m

Acquire.com reports a 3.9 times median profit multiple for its SaaS transactions in 2024 and 2025 and discusses a broad 3 to 5 times range. This is a global marketplace reference, not a Saudi edtech comparable or an appraisal. Its profit metric must be reconciled to your earnings measure; it should not be relabelled EV/EBITDA without justification. [S23]

A negotiated enterprise value, equity value, cash received at closing and founder proceeds can differ because of debt, cash, working capital, ownership percentage, taxes, fees and earnouts. Do not promise a Vision 2030 valuation premium. Strategic fit matters only when a real buyer is willing to pay for it.

39 Sale process and immediate next steps

Prepare a transferable company

- Build 12 months of consistent financial and cohort reporting where feasible. Reconcile claimed revenue and profit to bank and payment records.
- Reduce founder dependence with deployment, content update, support and incident runbooks. Measure what replacement staffing costs.
- Document your original character/media rights and software ownership. Using open model weights does not mean owning those foundation models exclusively.
- Identify potential strategic buyers by fit: training groups, workforce learning providers, enterprise education platforms and companies needing Arabic practical AI training. These are buyer categories, not expressions of interest.
- Prepare a short teaser, evidence backed management presentation and organised data room. Use professional advice on transaction structure and disclosure.
- Compare offers by cash at closing, contingent payments, liabilities, retained responsibilities and restrictions. A headline price alone can be misleading.

Your next ten working actions

Order	Concrete action	Finish condition
1	Choose the original character and voice rights.	Asset sheet and consent/licence record.
2	Try the delivered prototype and confirm the one lab task.	One learner completes it without explanation from you.
3	Run the backend starter locally in deterministic mode.	Tests and exported artifact succeed.
4	Enable real model drafts and the browser adapter.	Saved output reflects actual tool execution.
5	Produce the eight minute recorded lesson.	Reviewed MP4, captions and assignment.
6	Connect speech and the custom avatar worker.	Real ten minute session with measured timing.
7	Run the acceptance set and cost logging.	Failures recorded and release decision made.
8	Produce the remaining lessons in batches.	Each week has working labs and assessments.
9	Complete paid launch applicability and payment setup.	Documented approvals/process and purchase test.
10	Run a small paid cohort and review outcomes.	Completion, price, support and retention evidence.

The next purchase should resolve a measured blocker, not anticipate a 20 course organisation before the first lesson works. Keep the original human, verified screen actions and a useful learner outcome as the centre of the build.

40 Implement the real voice and avatar path

Build the layers in this order so each experiment has a clear result. Keep the provided deterministic backend available as a regression mode, and keep the real provider mode explicit. No foundation model training is required for this starting architecture.

Task	Implementation steps	Finish condition
Connect the product UI	Replace scripted API calls with the backend session, message, action and state contracts. Keep server credentials outside the browser and map error states into the interface.	A real draft, source reference and saved artifact appear in the polished UI.
Add Arabic listening	Capture push to talk audio; resample to the ASR input; transcribe with a multilingual Whisper model through whisper.cpp. Send final text into the existing message endpoint.	Twenty representative utterances have recorded results and errors.
Add Arabic speaking	Run Chatterbox Multilingual V3 in an isolated speech worker. Select its Arabic path and owned voice reference; normalise tool names and numbers. Return audio metadata and duration.	The tutor reads ten representative explanations clearly.
Prepare the presenter	Produce a clean original base clip and preprocess face assets for MuseTalk. Test approved audio against it before live integration.	A stable five scene clip with checked sync.
Build the streaming adapter	Receive short audio segments; preserve rolling context; render frames; yield synchronized audio/video frames through the LiveKit generator interface; carry timestamps in the worker protocol; bound queues and clear them on epoch changes.	Sustained measured playback, valid timing and working interruption.
Publish the lab screen	Capture the dedicated browser display and publish it as its own room track. Keep model observations and typed actions on separate APIs.	The learner sees the same state that the action verifier reads.
Join the full loop	Connect learner speech, retrieval, tool decisions, verified observations, explanation and face playback. Run cancellation and disconnect cases.	The ten minute real demo passes its acceptance gates.

Do not combine MuseTalk and Chatterbox dependency environments blindly. Their documented runtimes differ. Package them as separate workers with an explicit audio protocol and model versions; measure any GPU sharing before including it in capacity assumptions. [S05–S06]

41 Deploy a reusable virtual lab

Use the local browser until the lesson works. The first hosted lab should be a dedicated browser service with a small resource limit. A desktop VM is an additional template for courses that genuinely require desktop applications; it is not the same thing as an incognito browser context.

Step	Implementation detail
1 Define the template	Create a manifest with template_id, version, CPU/RAM limits, browser/app versions, entry URL, fixture, allowed tools, network destinations and maximum session duration.
2 Build the environment	Package the controlled web lab and Playwright dependencies in a tested Linux image. Install the matching Chromium binary and needed fonts. For a real desktop app, build a separately licensed VM image with a desktop session and capture service.
3 Add the lab API	Expose health, read state, execute typed action, screenshot, export and reset operations to the coordinator. Keep them inaccessible to arbitrary internet callers.
4 Add screen publication	Capture only the learner lab display. Publish to the assigned LiveKit room with a short lived token. Verify cursor visibility, readable resolution, Arabic fonts and reconnect behaviour.
5 Provision per session	After entitlement checks, create the lab, attach a fresh fixture, issue session credentials and health check. Return ready only when both execution and screen observation work.
6 Enforce limits	Set timeouts, resource quotas, allowlisted outbound traffic and a server side action policy. Do not mount a Docker socket, host filesystem or production secrets into a teaching lab.
7 Preserve and dispose	Save permitted artifacts to learner scoped storage. Revoke credentials and destroy the ephemeral lab. Test stale URLs, resource cleanup and the next learner’s clean state.
8 Scale by measurements	Measure startup time, memory and active session load. Add a queue and a small warm pool; autoscale workers using actual demand. Keep expensive avatar and video generation resources separately metered.

For recorded production, start the same template with a fixed fixture, execute the recipe, capture the verified run and save its manifest. For live teaching, start a fresh instance for each session and allow learner takeover. Reusing templates and recipes does not mean reusing a previous learner’s open session.

The backend starter currently supplies the local lab adapter and screenshots. Hosted template provisioning, VM images, screen streaming and this deployment lifecycle remain implementation tasks. Use the selected provider documentation and regional requirements when choosing the host.

References

Primary technical, pricing and regulatory sources checked on 23 October 2026. Forecasts, targets, budgets, course design and valuation scenarios are planning recommendations; vendor documentation does not establish achieved product performance.

S01 Google Gemini 3.5 Flash Lite model specification

https://ai.google.dev/gemini-api/docs/models/gemini-3.5-flash-lite

S02 Playwright browser context isolation

https://playwright.dev/docs/browser-contexts

S03 FLUX.1 schnell model card

https://huggingface.co/black-forest-labs/FLUX.1-schnell

S04 Wan 2.2 TI2V 5B model card

https://huggingface.co/Wan-AI/Wan2.2-TI2V-5B

S05 Chatterbox official repository

https://github.com/resemble-ai/chatterbox

S06 MuseTalk official repository

https://github.com/TMElyralab/MuseTalk

S07 whisper.cpp official repository

https://github.com/ggml-org/whisper.cpp

S08 OBS Studio

https://obsproject.com/

S09 LiveKit virtual avatars and custom workers

https://docs.livekit.io/agents/models/avatar/

S10 LiveKit Python avatar API reference

https://docs.livekit.io/reference/python/livekit/agents/voice/avatar/

S11 MuseTalk realtime inference implementation

https://github.com/TMElyralab/MuseTalk/blob/main/scripts/realtime_inference.py

S12 Runpod A6000 published rates

https://www.runpod.io/gpu-models/rtx-a6000

References continued

Primary technical, pricing and regulatory sources checked on 23 October 2026. Forecasts, targets, budgets, course design and valuation scenarios are planning recommendations; vendor documentation does not establish achieved product performance.

S13 Runpod Pod and storage pricing

https://docs.runpod.io/pods/pricing

S14 E2B sandbox pricing and plan limits

https://e2b.dev/pricing

S15 Cloudflare Stream pricing

https://developers.cloudflare.com/stream/pricing/

S16 Gemini API pricing

https://ai.google.dev/gemini-api/docs/pricing

S17 LiveKit pricing

https://livekit.com/pricing

S18 NeLC unified private sector e learning licensing announcement

https://nelc.gov.sa/en/media-center/news/unified-e-learning-licensing-private-sector-en

S19 NeLC e learning regulations

https://nelc.gov.sa/en/regulations-and-standards/elearning-regulation

S20 SDAIA laws regulations and data transfer guidance

https://sdaia.gov.sa/en/SDAIA/about/Pages/RegulationsAndPolicies.aspx

S21 ZATCA VAT registration for individuals

https://zatca.gov.sa/en/eServices/Pages/eServices-002.aspx

S22 ZATCA e invoicing introduction and applicability

https://zatca.gov.sa/en/E-Invoicing/Introduction/Pages/What-is-e-invoicing.aspx

S23 Acquire.com SaaS valuation multiples discussion

https://blog.acquire.com/saas-valuation-multiples/

Recheck before committing

Confirm current model identifiers, licences, provider region availability, GPU capacity, prices and Saudi requirements when you implement. Store the exact versions and commercial terms used in the repository and audit register. The cost workbook remains the editable record of assumptions behind the figures in this specification.